



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



Direct-to-IP Malware C2 and DNS Bypass Detection Priorities

Threat Report

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TLP:CLEAR | Lost Boys Cyber | Direct-to-IP Malware C2 and DNS Bypass Detection Priorities - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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Direct-to-IP Malware C2 and DNS Bypass Detection Priorities

1. Executive Summary

This report was selected for the 2026-09-03 UTC Lost Boys Cyber daily coverage set because it presents actionable defender relevance, current source coverage, and clear opportunities for detection or threat hunting. The activity or trend should be reviewed by SOC, vulnerability management, identity, and sector-risk owners.

Primary exposure: Organizations relying heavily on DNS security controls without equally strong egress and proxy telemetry. The report does not assert victim compromise beyond public source claims; it converts available reporting into defensive priorities, hunt hypotheses, and operational controls.

2. Intelligence Requirements

Question	Current Answer	Confidence
What happened?	Public reporting describes Direct-to-IP Malware C2 and DNS Bypass Detection Priorities.	High
Who should care?	Organizations relying heavily on DNS security controls without equally strong egress and proxy telemetry.	High
What can defenders do today?	Review exposure, run the included hunts, and validate mitigations against recent telemetry.	High

3. Source Review & Confidence

Source	Contribution	Confidence
Unit 42 direct-to-IP malware C2	Analysis of 4 million dynamic-analysis reports found 45.32% of malware samples with C2 used direct-to-IP connections.	Medium-High
Unit 42 threat research index	Context for current Unit 42 research and incident-response findings.	Medium-High
M-Trends 2026	Frontline incident data for attacker persistence and detection priorities.	Medium-High

Source depth was sufficient for a defensive threat report. Hard IOCs were not consistently available in collected public sources, so this report emphasizes TTP and telemetry-based analytics.

4. Threat Activity Overview

Reporting points to Multiple malware families and intrusion operators activity/trends involving Application-layer C2, Direct IP egress, Proxy bypass, Defense evasion against DNS-only monitoring. For defenders, the highest-value action is to convert these observations into exposure review and telemetry hunts rather than waiting for environment-specific IOCs.

5. Affected Sectors and Exposure

Exposure Area	Why It Matters	Recommended Review
Identity and remote access	Valid accounts reduce attacker friction and bypass many perimeter controls.	Review MFA fatigue, impossible travel, help-desk reset and risky sign-in patterns.

Internet-facing applications	Public services remain common initial access and implant placement points.	Confirm patch status and inspect web logs for unusual POST/URI/user-agent patterns.
Endpoint and developer workstations	Lures and developer workflows can execute scripts or malware locally.	Monitor Node.js, PowerShell, shell and archive execution from user-writable paths.
Third-party and shared platforms	Aviation and other operators depend on vendors and SaaS that can become blast-radius multipliers.	Review vendor access, SSO integrations, and contingency procedures.

6. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	T1566 Phishing	Application-layer C2 described in source reporting or analyst synthesis.
Execution/C2/Impact	T1059 Command and Scripting Interpreter	Direct IP egress described in source reporting or analyst synthesis.
Execution/C2/Impact	T1071 Application Layer Protocol	Proxy bypass described in source reporting or analyst synthesis.
Execution/C2/Impact	T1078 Valid Accounts	Defense evasion against DNS-only monitoring described in source reporting or analyst synthesis.

7. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
No normalized hard IOCs available in collected public source set	Source limitation	Hunt on behaviors and telemetry patterns instead of relying on blocklists.
Cloud-hosted C2 / direct-IP / SaaS identity patterns where applicable	Behavioral infrastructure	Review egress, proxy, Entra and EDR logs for suspicious flows.

8. Detection Engineering

This section includes deployable starting points. Tune filters to local baselines before production alerting.

8.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious scripting or developer-runtime execution from downloads/temp paths	EDR process creation	Catch lure-driven malware execution.
Direct-to-IP or rare ASN egress from user endpoints	Network/EDR	Catch DNS-bypass C2 and unusual beaconing.
Risky cloud identity sign-ins and MFA reset patterns	Entra/IdP logs	Catch credential-theft and social-engineering outcomes.

8.2. Sigma / Detection Content

title: Direct-to-IP Malware C2 and DNS Bypass Detection Priorities - Suspicious Activity Heuristic id: auto-generated-direct-to-ip-malware-c2-and-dns-bypass-detection

```

status: experimental
description: Detects process or network command-line clues that should be reviewed in context for this report.
logsource:
  product: windows
  category: process_creation
detection:
  selection_cmd:
    CommandLine|contains:
      - 'Multiple malware families and intrusion operators'
      - 'Application-layer C2'
      - 'Direct IP egress'
      - 'Proxy bypass'
  condition: selection_cmd
falsepositives:
  - Administrative validation, vulnerability scanning, or approved red-team testing.
level: medium

```

8.3. Hunt Query

```

// Hunt for suspicious activity related to Direct-to-IP Malware C2 and DNS Bypass Detection Priorities
DeviceNetworkEvents
| where Timestamp > ago(14d)
| where RemoteUrl has_any ('Multiple malware families and intrusion operators', 'Application-layer C2',
'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only monitoring') or
InitiatingProcessCommandLine has_any ('Multiple malware families and intrusion operators',
'Application-layer C2', 'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only
monitoring')
| project Timestamp, DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName, RemoteUrl,
RemoteIP, InitiatingProcessCommandLine

```

9. Threat Hunting

9.1. Hunt Hypothesis

If this activity is present, telemetry should show a cluster of suspicious user execution, rare network egress, risky sign-in behavior, or web-server process anomalies connected by account, device, or time window.

9.2. Hunt Query

```

let lookback = 14d;
let suspect_terms = dynamic(['Multiple malware families and intrusion operators', 'Application-layer
C2', 'Direct IP egress', 'Proxy bypass', 'Defense evasion against DNS-only monitoring']);
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where ProcessCommandLine has_any (suspect_terms) or FolderPath has_any ("\\Downloads\\", "\\Temp\\",
"/tmp/", "/var/www/")
| summarize FirstSeen=min(Timestamp), LastSeen=max(Timestamp), Cmds=makeset(ProcessCommandLine, 10) by
DeviceName, InitiatingProcessAccountName, FileName
| order by LastSeen desc

```

10. Risk Assessment

Dimension	Rating	Rationale
Likelihood	Medium	Current public reporting and common exposure paths make opportunistic contact plausible.
Impact	High	Credential theft, malware execution, or shared-platform compromise can produce operational disruption.
Detection Maturity Need	High	Behavioral analytics are required because

		hard IOCs may be absent or short-lived.
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11. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the included KQL hunts and review hits with identity and endpoint context.
High	Vulnerability / Platform	Confirm public-service patching, exposed admin interfaces, and third-party access restrictions.
High	Identity	Enforce phishing-resistant MFA for privileged and developer accounts; review risky sign-ins.
Medium	Resilience	Update incident runbooks for third-party/shared-platform disruption.

12. References

- [1] Unit 42 direct-to-IP malware C2: <https://unit42.paloaltonetworks.com/malware-bypass-dns-direct-to-ip/>
- [2] Unit 42 threat research index: <https://unit42.paloaltonetworks.com/>
- [3] M-Trends 2026: <https://cloud.google.com/blog/topics/threat-intelligence/m-trends-2026>